公式推导

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1　两球三弹簧谐振子

如图1，该物理模型由两个完全相同的小球（质量为 m）与三个弹簧（劲度系数 $k = m\omega^2$）连接，以及两面质量无穷大的刚体墙构成。

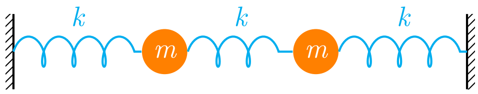

图 1: 两球三弹簧谐振子

该系统的拉格朗日量如下表示。

$$\begin{cases} L = T - V \\ T = \dfrac{1}{2}m\dot{x}_1^2 + \dfrac{1}{2}m\dot{x}_2^2 \\ V = \dfrac{1}{2}m\omega^2 x_1^2 + \dfrac{1}{2}m\omega^2 x_2^2 + \dfrac{1}{2}m\omega^2\left(x_1 - x_2\right)^2 \end{cases} \tag{1}$$

此时有

$$\begin{cases} \dfrac{\partial}{\partial x_1}L - \dfrac{d}{dt}\left(\dfrac{\partial}{\partial \dot{x}_1}L\right) = 0 \\ \dfrac{\partial}{\partial x_2}L - \dfrac{d}{dt}\left(\dfrac{\partial}{\partial \dot{x}_2}L\right) = 0 \end{cases} \implies \frac{\mathrm{d}^2}{\mathrm{d}t^2}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \omega^2 \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \tag{2}$$

对角化即有

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\begin{pmatrix} \dfrac{1}{\sqrt{2}}(x_1 + x_2) \\ \dfrac{1}{\sqrt{2}}(x_1 - x_2) \end{pmatrix} + \omega^2 \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}\begin{pmatrix} \dfrac{1}{\sqrt{2}}(x_1 + x_2) \\ \dfrac{1}{\sqrt{2}}(x_1 - x_2) \end{pmatrix} = 0 \tag{3}$$

即有

$$x_1 + x_2 = C_{\mathrm{c}1}\cos(\omega t) + C_{\mathrm{s}1}\sin(\omega t) \tag{4}$$

$$x_1 - x_2 = C_{\mathrm{c}3}\cos(\sqrt{3}\omega t) + C_{\mathrm{s}3}\sin(\sqrt{3}\omega t) \tag{5}$$

其中 $C_{\mathrm{c}1}$、$C_{\mathrm{s}1}$、$C_{\mathrm{c}3}$、$C_{\mathrm{s}3}$ 均为与初始状态相匹配的常数。
初始状态与常数的关系如下

$$\begin{cases} x_1(t=0) = X_1 \\ x_2(t=0) = X_2 \\ \dot{x}_1(t=0) = V_1 \\ \dot{x}_2(t=0) = V_2 \end{cases} \implies \begin{cases} C_{\mathrm{c}1} = X_1 + X_2 \\ C_{\mathrm{c}3} = X_1 - X_2 \\ C_{\mathrm{c}1} = \dfrac{V_1 + V_2}{\omega} \\ C_{\mathrm{c}3} = \dfrac{V_1 - V_2}{\sqrt{3}\omega} \end{cases} \tag{6}$$

2 双摆系统

如图2，两个完全相同的小球通过长度相同的不可延伸硬杆相连接，构成双摆。该系统具备混沌的特点，但并非所有初态均对应混沌状态。

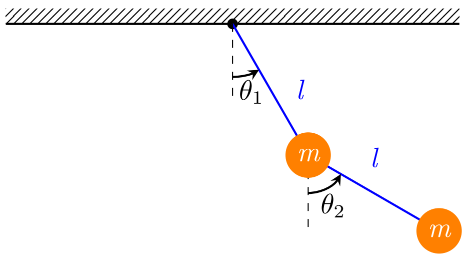

图 2: 双摆系统

2.1 物理过程

该体系拉格朗日量如下所示

$$\begin{cases} L = T - V \\ T = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 + \frac{1}{2}m\dot{y}_1^2 + \frac{1}{2}m\dot{y}_2^2 \\ V = -mgy_1 - mgy_2 \end{cases} \tag{7}$$

根据双摆的规则，有

$$\begin{cases} x_1 = l\sin(\theta_1) \\ x_2 = l\sin(\theta_1) + l\sin(\theta_2) \\ y_1 = l\cos(\theta_1) \\ y_2 = l\cos(\theta_1) + l\cos(\theta_2) \end{cases} \implies \begin{cases} \dot{x}_1 = l\cos(\theta_1)\dot{\theta}_1 \\ \dot{x}_2 = l\cos(\theta_1)\dot{\theta}_1 + l\cos(\theta_2)\dot{\theta}_2 \\ \dot{y}_1 = -l\sin(\theta_1)\dot{\theta}_1 \\ \dot{y}_2 = -l\sin(\theta_1)\dot{\theta}_1 - l\sin(\theta_2)\dot{\theta}_2 \end{cases} \tag{8}$$

代入则有

$$\begin{cases} T = \frac{1}{2}ml^2\left(\cos^2(\theta_1)\dot{\theta}_1^2 + \left(\cos(\theta_1)\dot{\theta}_1 + \cos(\theta_2)\dot{\theta}_2\right)^2 + \sin^2(\theta_1)\dot{\theta}_1^2 + \left(\sin(\theta_1)\dot{\theta}_1 + \sin(\theta_2)\dot{\theta}_2\right)^2\right) \\ V = -2mgl\cos(\theta_1) - mgl\cos(\theta_2) \end{cases} \tag{9}$$

即

$$\begin{cases} T = \frac{1}{2}ml^2\left(2\dot{\theta}_1^2 + \dot{\theta}_2^2 + 2\cos(\theta_1 - \theta_2)\dot{\theta}_1\dot{\theta}_2\right) \\ V = -2mgl\cos(\theta_1) - mgl\cos(\theta_2) \end{cases} \tag{10}$$

$$L = \frac{1}{2}ml^2\left(2\dot{\theta}_1^2 + \dot{\theta}_2^2 + 2\cos(\theta_1 - \theta_2)\dot{\theta}_1\dot{\theta}_2\right) + 2mgl\cos(\theta_1) + mgl\cos(\theta_2) \tag{11}$$

固有

$$\frac{\partial L}{\partial \theta_1} = \frac{1}{2}ml^2 \cdot -2\sin(\theta_1-\theta_2)\dot\theta_1\dot\theta_2 - 2mgl\sin(\theta_1) \tag{12}$$

$$\frac{\partial L}{\partial \theta_2} = \frac{1}{2}ml^2 \cdot 2\sin(\theta_1-\theta_2)\dot\theta_1\dot\theta_2 - mgl\sin(\theta_2) \tag{13}$$

$$\frac{\partial L}{\partial \dot\theta_1} = \frac{1}{2}ml^2\left(4\dot\theta_1 + 2\cos(\theta_1-\theta_2)\dot\theta_2\right) \tag{14}$$

$$\frac{\partial L}{\partial \dot\theta_2} = \frac{1}{2}ml^2\left(2\dot\theta_2 + 2\cos(\theta_1-\theta_2)\dot\theta_1\right) \tag{15}$$

则

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot\theta_1}\right) = \frac{1}{2}ml^2\left(4\ddot\theta_1 + 2\cos(\theta_1-\theta_2)\ddot\theta_2 - 2\sin(\theta_1-\theta_2)\dot\theta_1\dot\theta_2 + 2\sin(\theta_1-\theta_2)\dot\theta_2^2\right) \tag{16}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot\theta_2}\right) = \frac{1}{2}ml^2\left(2\ddot\theta_2 + 2\cos(\theta_1-\theta_2)\ddot\theta_1 + 2\sin(\theta_1-\theta_2)\dot\theta_1\dot\theta_2 - 2\sin(\theta_1-\theta_2)\dot\theta_1^2\right) \tag{17}$$

代入

$$\begin{cases} \dfrac{\partial}{\partial \theta_1}L = \dfrac{d}{dt}\left(\dfrac{\partial}{\partial \dot\theta_1}L\right) \\ \dfrac{\partial}{\partial \theta_2}L = \dfrac{d}{dt}\left(\dfrac{\partial}{\partial \dot\theta_2}L\right) \end{cases} \tag{18}$$

有

$$\begin{cases} \dfrac{1}{2}ml^2\left(4\ddot\theta_1 + 2\cos(\theta_1-\theta_2)\ddot\theta_2 + 2\sin(\theta_1-\theta_2)\dot\theta_2^2\right) + 2mgl\sin(\theta_1) = 0 \\ \dfrac{1}{2}ml^2\left(2\ddot\theta_2 + 2\cos(\theta_1-\theta_2)\ddot\theta_1 - 2\sin(\theta_1-\theta_2)\dot\theta_1^2\right) + mgl\sin(\theta_2) = 0 \end{cases} \tag{19}$$

考虑 $\omega = \sqrt{\dfrac{g}{l}}$，有

$$\begin{cases} 2\ddot\theta_1 + \cos(\theta_1-\theta_2)\ddot\theta_2 + \sin(\theta_1-\theta_2)\dot\theta_2^2 + 2\omega^2\sin(\theta_1) = 0 \\ \ddot\theta_2 + \cos(\theta_1-\theta_2)\ddot\theta_1 - \sin(\theta_1-\theta_2)\dot\theta_1^2 + \omega^2\sin(\theta_2) = 0 \end{cases} \tag{20}$$

2.2 数值离散化

考虑数值微分运算

$$
\begin{cases}
\dfrac{\mathrm{d}y}{\mathrm{d}x} \approx \dfrac{y(x+\Delta x) - y(x-\Delta x)}{2\Delta x} \\
\dfrac{\mathrm{d}^2 y}{\mathrm{d}x^2} \approx \dfrac{y(x+\Delta x) + y(x-\Delta x) - 2y(x)}{(\Delta x)^2}
\end{cases}
\tag{21}
$$

令 $\theta_i^{(n)}$ 为第 n 次迭代后 θ_i 的值，有

$$
\begin{cases}
\dot{\theta}_1 \approx \dfrac{\theta_1^{(n+1)} - \theta_1^{(n-1)}}{2\Delta t} \\
\dot{\theta}_2 \approx \dfrac{\theta_2^{(n+1)} - \theta_2^{(n-1)}}{2\Delta t} \\
\ddot{\theta}_1 \approx \dfrac{\theta_1^{(n+1)} + \theta_1^{(n-1)} - 2\theta_1^{(n)}}{(\Delta t)^2} \\
\ddot{\theta}_2 \approx \dfrac{\theta_2^{(n+1)} + \theta_2^{(n-1)} - 2\theta_2^{(n)}}{(\Delta t)^2}
\end{cases}
\tag{22}
$$

令

$$
\begin{cases}
C_{12} = \cos(\theta_1 - \theta_2) \\
S_{12} = \sin(\theta_1 - \theta_2)
\end{cases}
\tag{23}
$$

回代方程20则有

$$
\begin{cases}
2\dfrac{\theta_1^{(n+1)} + \theta_1^{(n-1)} - 2\theta_1^{(n)}}{(\Delta t)^2} + C_{12}^{(n)}\dfrac{\theta_2^{(n+1)} + \theta_2^{(n-1)} - 2\theta_2^{(n)}}{(\Delta t)^2} + S_{12}^{(n)}\left(\dfrac{\theta_2^{(n+1)} - \theta_2^{(n-1)}}{2\Delta t}\right)^2 + 2\omega^2\sin\left(\theta_1^{(n)}\right) = 0 \\
\dfrac{\theta_2^{(n+1)} + \theta_2^{(n-1)} - 2\theta_2^{(n)}}{(\Delta t)^2} + C_{12}^{(n)}\dfrac{\theta_1^{(n+1)} + \theta_1^{(n-1)} - 2\theta_1^{(n)}}{(\Delta t)^2} - S_{12}^{(n)}\left(\dfrac{\theta_1^{(n+1)} - \theta_1^{(n-1)}}{2\Delta t}\right)^2 + \omega^2\sin\left(\theta_2^{(n)}\right) = 0
\end{cases}
$$

即

$$
\begin{bmatrix}
0 & S_{12}^{(n)} & 8 & 4C_{12}^{(n)} - 2S_{12}^{(n)}\theta_2^{(n-1)} & c_1 \\
-S_{12}^{(n)} & 0 & 4C_{12}^{(n)} + 2S_{12}^{(n)}\theta_1^{(n-1)} & 4 & c_2
\end{bmatrix}
\begin{pmatrix}
\left(\theta_1^{(n+1)}\right)^2 \\
\left(\theta_2^{(n+1)}\right)^2 \\
\theta_1^{(n+1)} \\
\theta_2^{(n+1)} \\
1
\end{pmatrix}
= 0
\tag{24}
$$

其中

$$
\begin{cases}
c_1 = 8\left(\theta_1^{(n-1)} - 2\theta_1^{(n)}\right) + 4C_{12}^{(n)}\left(\theta_2^{(n-1)} - 2\theta_2^{(n)}\right) + S_{12}^{(n)}\left(\theta_2^{(n-1)}\right)^2 + 8\left(\omega\Delta t\right)^2\sin\left(\theta_1^{(n)}\right) \\
c_2 = 4\left(\theta_2^{(n-1)} - 2\theta_2^{(n)}\right) + 4C_{12}^{(n)}\left(\theta_1^{(n-1)} - 2\theta_1^{(n)}\right) - S_{12}^{(n)}\left(\theta_1^{(n-1)}\right)^2 + 4\left(\omega\Delta t\right)^2\sin\left(\theta_2^{(n)}\right)
\end{cases}
\tag{25}
$$

该方程需要根据 $S_{12}^{(n)} = 0$ 是否成立分情况讨论。

2.2.1 若考虑二次项系数为零

即 $S_{12}^{(n)}=0$，此时有 $\theta_1^{(n)}=\theta_2^{(n)}+k\cdot\pi, k\in\mathbb{Z}$，此时方程退化为

$$\begin{bmatrix} 8 & 4C_{12}^{(n)} \\ 4C_{12}^{(n)} & 4 \end{bmatrix}\begin{pmatrix} \theta_1^{(n+1)} \\ \theta_2^{(n+1)} \end{pmatrix}=\begin{pmatrix} -c_1 \\ -c_2 \end{pmatrix} \tag{26}$$

其中

$$\begin{cases} C_{12}^{(n)}=\pm 1 \\ \left(C_{12}^{(n)}\right)^2=1 \\ \sin\left(\theta_2^{(n)}\right)=C_{12}^{(n)}\sin\left(\theta_1^{(n)}\right) \\ \sin\left(\theta_1^{(n)}\right)=C_{12}^{(n)}\sin\left(\theta_2^{(n)}\right) \\ c_1=8\left(\theta_1^{(n-1)}-2\theta_1^{(n)}\right)+4C_{12}^{(n)}\left(\theta_2^{(n-1)}-2\theta_2^{(n)}\right)+8\left(\omega\Delta t\right)^2\sin\left(\theta_1^{(n)}\right) \\ c_2=4\left(\theta_2^{(n-1)}-2\theta_2^{(n)}\right)+4C_{12}^{(n)}\left(\theta_1^{(n-1)}-2\theta_1^{(n)}\right)+4\left(\omega\Delta t\right)^2\sin\left(\theta_2^{(n)}\right) \end{cases} \tag{27}$$

解得

$$\begin{cases} \theta_1^{(n+1)} & =\dfrac{1}{4}(C_{12}^{(n)}c_2-c_1) \\ \theta_2^{(n+1)} & =\dfrac{1}{4}(C_{12}^{(n)}c_1-2c_2) \end{cases} \tag{28}$$

即

$$\begin{cases} \theta_1^{(n+1)} & =\left(\left(2\theta_1^{(n)}-\theta_1^{(n-1)}\right)+C_{12}^{(n)}\left(\omega\Delta t\right)^2\sin\left(\theta_2^{(n)}\right)-2\left(\omega\Delta t\right)^2\sin\left(\theta_1^{(n)}\right)\right) \\ \theta_2^{(n+1)} & =\left(\left(2\theta_2^{(n)}-\theta_2^{(n-1)}\right)-2\left(\omega\Delta t\right)^2\sin\left(\theta_2^{(n)}\right)+2C_{12}^{(n)}\left(\omega\Delta t\right)^2\sin\left(\theta_1^{(n)}\right)\right) \end{cases} \tag{29}$$

2.2.2 考虑二次项系数不为零

即 $S_{12}^{(n)} \neq 0$，定义系数

$$\begin{cases} a & = S_{12}^{(n)} \\ b_{11} & = \theta_1^{(n-1)} + 2\cot(\theta_1^{(n)} - \theta_2^{(n)}) \\ b_{12} & = \frac{2}{S_{12}^{(n)}} \\ b_{21} & = -\frac{4}{S_{12}^{(n)}} \\ b_{22} & = \theta_2^{(n-1)} - 2\cot(\theta_1^{(n)} - \theta_2^{(n)}) \\ c_1 & = 8\left(\theta_1^{(n-1)} - 2\theta_1^{(n)}\right) + 4C_{12}^{(n)}\left(\theta_2^{(n-1)} - 2\theta_2^{(n)}\right) + S_{12}^{(n)}\left(\theta_2^{(n-1)}\right)^2 + 8\left(\omega\Delta t\right)^2\sin\left(\theta_1^{(n)}\right) \\ c_2 & = 4\left(\theta_2^{(n-1)} - 2\theta_2^{(n)}\right) + 4C_{12}^{(n)}\left(\theta_1^{(n-1)} - 2\theta_1^{(n)}\right) - S_{12}^{(n)}\left(\theta_1^{(n-1)}\right)^2 + 4\left(\omega\Delta t\right)^2\sin\left(\theta_2^{(n)}\right) \end{cases} \tag{30}$$

对于该方程则有

$$\begin{bmatrix} a & 0 & -2ab_{11} & -2ab_{12} & -c_2 \\ 0 & a & -2ab_{21} & -2ab_{22} & c_1 \end{bmatrix} \begin{pmatrix} \left(\theta_1^{(n+1)}\right)^2 \\ \left(\theta_2^{(n+1)}\right)^2 \\ \theta_1^{(n+1)} \\ \theta_2^{(n+1)} \\ 1 \end{pmatrix} = 0 \tag{31}$$

考虑

$$\begin{cases} \gamma_1 & = -\dfrac{c_2}{a} - b_{11}^2 - 2b_{12}b_{22} \\ \gamma_2 & = \dfrac{c_1}{a} - b_{22}^2 - 2b_{11}b_{21} \end{cases} \tag{32}$$

$$\begin{cases} \theta_{c1}^{(n+1)} = \theta_1^{(n+1)} - b_{11} = \theta_1^{(n+1)} - \left(\theta_1^{(n-1)} + 2\cot(\theta_1^{(n)} - \theta_2^{(n)})\right) \\ \theta_{c2}^{(n+1)} = \theta_2^{(n+1)} - b_{22} = \theta_2^{(n+1)} - \left(\theta_2^{(n-1)} - 2\cot(\theta_1^{(n)} - \theta_2^{(n)})\right) \end{cases} \tag{33}$$

即有

$$\begin{bmatrix} 1 & 0 & 0 & -2b_{12} & \gamma_1 \\ 0 & 1 & -2b_{21} & 0 & \gamma_2 \end{bmatrix} \begin{pmatrix} \left(\theta_{c1}^{(n+1)}\right)^2 \\ \left(\theta_{c2}^{(n+1)}\right)^2 \\ \theta_{c1}^{(n+1)} \\ \theta_{c2}^{(n+1)} \\ 1 \end{pmatrix} = 0 \tag{34}$$

其中 $b_{12} \neq 0$ 与 $b_{21} \neq 0$ 恒成立。通过消元可得

$$\begin{cases} (\theta_{c1}^{(n+1)})^4 + 2\gamma_1(\theta_{c1}^{(n+1)})^2 - 8b_{12}^2 b_{21}\theta_{c1}^{(n+1)} + (4b_{12}^2\gamma_2 + \gamma_1^2) = 0 \\ \theta_{c2}^{(n+1)} = \dfrac{(\theta_{c1}^{(n+1)})^2 + \gamma_1}{2b_{12}} \end{cases} \tag{35}$$

注意到 $b_{21} \equiv -2b_{12}$，令 $b_{12} = b$，并换元

$$\begin{cases} p = 2\gamma_1 \\ q = -8b_{12}^2 b_{21} = 16b^3 \\ r = (4b^2\gamma_2 + \gamma_1^2) \end{cases} \tag{36}$$

易得 $\theta_{c1}^{(n+1)}$ 是方程37的其中一根。

$$x^4 + px^2 + qx + r = 0 \tag{37}$$

考虑将该方程转化为的形式

$$(x^2 + \alpha)^2 - (\sqrt{2\alpha - p}x - \frac{q}{2\sqrt{2\alpha - p}})^2 = 0 \tag{38}$$

则系数 α 需要满足

$$\alpha^2 - \frac{q^2}{4(2\alpha - p)} = r \Longrightarrow 8\alpha^3 - 4p\alpha^2 - 8r\alpha + 4pr - q^2 = 0 \tag{39}$$

而方程的根在复数域则为（$2\alpha - p < 0$ 同理，此处不讨论 $\sqrt{-1}$ 与 $\pm\mathrm{i}$ 问题）

$$\begin{cases} x_1 & = \frac{1}{2}\sqrt{2\alpha - p} + \frac{1}{2}\sqrt{-2\alpha - p - \frac{2q}{\sqrt{2\alpha - p}}} \\ x_2 & = \frac{1}{2}\sqrt{2\alpha - p} - \frac{1}{2}\sqrt{-2\alpha - p - \frac{2q}{\sqrt{2\alpha - p}}} \\ x_3 & = -\frac{1}{2}\sqrt{2\alpha - p} + \frac{1}{2}\sqrt{-2\alpha - p + \frac{2q}{\sqrt{2\alpha - p}}} \\ x_4 & = -\frac{1}{2}\sqrt{2\alpha - p} - \frac{1}{2}\sqrt{-2\alpha - p + \frac{2q}{\sqrt{2\alpha - p}}} \end{cases} \tag{40}$$

考虑到这是该方程的 4 个不重复的根，考虑到顺序问题可以 α 的实际含义如下

$$\begin{cases} 2\alpha - p & = -(x_1 + x_2)(x_3 + x_4) \\ -\alpha - \frac{1}{2}p + \frac{1}{2}\sqrt{(2\alpha + p)^2 - \frac{4q^2}{2\alpha - p}} & = -(x_1 + x_3)(x_2 + x_4) \\ -\alpha - \frac{1}{2}p - \frac{1}{2}\sqrt{(2\alpha + p)^2 - \frac{4q^2}{2\alpha - p}} & = -(x_1 + x_4)(x_2 + x_3) \end{cases} \tag{41}$$

同一组解可以由 3 个不同的 α 生成，经验证这些 α 均为方程 39 的解。故仅需得到一个 α 便可求解出所有的根，故仅考虑方程 39 的实根。

综合方程35、36和方程35、40有

$$\begin{cases} \theta_{c1}^{(n+1)} = \pm_1 \sqrt{\frac{1}{2}(\alpha - \gamma_1)} \pm_2 \sqrt{-\frac{1}{2}(\alpha + \gamma_1) \mp_1 4b^3 \sqrt{\frac{2}{\alpha - \gamma_1}}} \\ \alpha^3 - \gamma_1\alpha^2 - r\alpha + (\gamma_1 r - 32b^6) = 0 \\ r = 4b^2\gamma_2 + \gamma_1^2 \end{cases} \tag{42}$$

正负号取决于解的具体形式。

对于 α ，令 $\alpha=\beta+\frac{1}{3}\gamma_1$ 易得

$$\beta^3-\frac{4}{3}\left(3b^2\gamma_2+\gamma_1^2\right)\beta+\frac{8}{27}\left(9\gamma_1\gamma_2b^2+2\gamma_1^3-108b^6\right)=0 \tag{43}$$

之前的中间参数 p、q 已不再使用，重新定义

$$\begin{cases}p=-\dfrac{4}{3}\left(3b^2\gamma_2+\gamma_1^2\right)\\ q=\dfrac{8}{27}\left(9\gamma_1\gamma_2b^2+2\gamma_1^3-108b^6\right)\end{cases} \tag{44}$$

考虑 $\beta=u+v$ 有

$$(u+v)^3+p(u+v)+q=0\Longrightarrow(u^3+v^3+q)+(3uv+p)(u+v)=0 \tag{45}$$

考虑特解，该解可以构造出一个满足条件 β

$$\begin{cases}u^3+v^3 &=-q\\ uv &=-\dfrac{1}{3}p\end{cases} \tag{46}$$

根据数形结合等方法可以判断 u^3 与 v^3 为方程的两根

$$x^2+qx-\frac{1}{27}p^3=0 \tag{47}$$

即（顺序无关，u、v 可以交换）

$$\begin{cases}u=\sqrt[3]{-\dfrac{1}{2}q+\sqrt{\dfrac{1}{27}p^3-\dfrac{1}{4}q^2}}=\dfrac{2}{3}\sqrt[3]{\dfrac{1}{2}\left(-(\ldots_q)+\sqrt{(\ldots_q)^2-4(\ldots_p)^3}\right)}\\ v=\sqrt[3]{-\dfrac{1}{2}q-\sqrt{\dfrac{1}{27}p^3-\dfrac{1}{4}q^2}}=\dfrac{2}{3}\sqrt[3]{\dfrac{1}{2}\left(-(\ldots_q)-\sqrt{(\ldots_q)^2-4(\ldots_p)^3}\right)}\end{cases} \tag{48}$$

联立方程30、33、35、42、43、48等易得

$$
\begin{cases}
\theta_1^{(n+1)} &= \theta_{c1}^{(n+1)} + \left(\theta_1^{(n-1)} + 2\cot(\theta_1^{(n)} - \theta_2^{(n)})\right) \\
\theta_2^{(n+1)} &= \theta_{c2}^{(n+1)} + \left(\theta_2^{(n-1)} - 2\cot(\theta_1^{(n)} - \theta_2^{(n)})\right) \\
\theta_{c2}^{(n+1)} &= \dfrac{\theta_{c1}^2 + \gamma_1}{2b} \\
\theta_{c1}^{(n+1)} &= \pm_1\sqrt{\dfrac{1}{2}(\alpha - \gamma_1)} \pm_2 \sqrt{-\dfrac{1}{2}(\alpha + \gamma_1) \mp_1 4b^3\sqrt{\dfrac{2}{\alpha - \gamma_1}}} \\
\alpha &= \frac{1}{3}\gamma_1 + u + v \\
u &= \dfrac{2}{3}\sqrt[3]{\dfrac{1}{2}\left(-\sigma_q + \sqrt{\sigma_q^2 - 4\sigma_p^3}\right)} \\
v &= \dfrac{2}{3}\sqrt[3]{\dfrac{1}{2}\left(-\sigma_q - \sqrt{\sigma_q^2 - 4\sigma_p^3}\right)} \\
\sigma_p &= 3b^2\gamma_2 + \gamma_1^2 \\
\sigma_q &= 9\gamma_1\gamma_2 b^2 + 2\gamma_1^3 - 108b^6 \\
\gamma_1 &= -\dfrac{c_2}{a} - b_{11}^2 - 2bb_{22} \\
\gamma_2 &= \dfrac{c_1}{a} - b_{22}^2 + 4bb_{11} \\
a &= \sin(\theta_1^{(n)} - \theta_2^{(n)}) \\
b &= 2\csc(\theta_1^{(n)} - \theta_2^{(n)}) \\
b_{11} &= \theta_1^{(n-1)} + 2\cot(\theta_1^{(n)} - \theta_2^{(n)}) \\
b_{22} &= \theta_2^{(n-1)} - 2\cot(\theta_1^{(n)} - \theta_2^{(n)}) \\
c_1 &= 8\left(\theta_1^{(n-1)} - 2\theta_1^{(n)}\right) + 4C_{12}^{(n)}\left(\theta_2^{(n-1)} - 2\theta_2^{(n)}\right) + S_{12}^{(n)}\left(\theta_2^{(n-1)}\right)^2 + 8\left(\omega\Delta t\right)^2\sin\left(\theta_1^{(n)}\right) \\
c_2 &= 4\left(\theta_2^{(n-1)} - 2\theta_2^{(n)}\right) + 4C_{12}^{(n)}\left(\theta_1^{(n-1)} - 2\theta_1^{(n)}\right) - S_{12}^{(n)}\left(\theta_1^{(n-1)}\right)^2 + 4\left(\omega\Delta t\right)^2\sin\left(\theta_2^{(n)}\right)
\end{cases}
\tag{49}
$$

在计算 u、v 与 $\theta_{c1}^{(n+1)}$ 时需要考虑虚数与分母为 0 情况

- $\sigma_q^2 - 4\sigma_p^3 < 0$ 的情况
  可以证明该情况等价于有 3 个实根的情况

$$
\alpha_1, \alpha_2, \alpha_3 = \frac{4}{3}\sqrt{\sigma_p}\cos\left(\frac{1}{3}\arccos\left(-\frac{\sigma_q}{2\sigma_p^{3/2}}\right) + \frac{2\pi k}{3}\right), \quad k = 0, 1, 2 \tag{50}
$$

- $\alpha - \gamma_1 = 0$ 的情况：过于复杂难以分析，在模拟过程中通过报错或纯数值解来计算

- $\alpha - \gamma_1 < 0$ 的情况：过于复杂难以分析，在模拟过程中通过报错或纯数值解来计算